

EE568 – Design of Electrical Machines

Lecture Notes

Middle East Technical University

Department of Electrical and Electronics Engineering

Compiled by Sefer İDACI

Spring 2026

Abstract

These notes were typeset from the handwritten course notes of EE568 “Design of Electrical Machines” at METU, Spring 2026. They follow the flow of the lectures: starting from magnetic-circuit fundamentals, then building up to the design of distributed and fractional-slot windings, to the output equation that sizes a machine, and finally to the practical constraints (mechanical, magnetic, thermal) that shape the final geometry. The classical references are the books by Pyrhönen et al. (*Design of Rotating Electrical Machines*, Wiley) and Hanselman (*Brushless Permanent Magnet Motor Design*, Magna Physics); equation numbering does not follow either book.

Contents

1	Magnetic Fundamentals	4
1.1	Electric–Magnetic Analogy	4
1.2	Air is a Circuit Element	4
1.3	B–H Curves and Permeability	4
1.4	Core Losses	5
1.5	Faraday’s Law and Inductance	5
1.6	Non-Uniform Flux, Fringing and Leakage	6
1.7	Energy Storage in the Magnetic Field	6
2	AC Machine Equivalent Circuit	7
3	Integral-Slot Distributed Windings	7
3.1	Slots per Pole per Phase	7
3.2	MMF of a Concentrated Coil	8

3.3	Distribution Factor k_d	8
3.4	Pitch Factor k_p	8
3.5	Winding Factor k_w	9
3.6	Worked Example – 180-slot, 50 Hz, 10-Pole Alternator	9
4	Fractional-Slot Windings	10
4.1	Star of Slots	10
4.2	Balancing Condition	11
4.3	Slot/Pole Combinations	11
4.4	Worked Example – 12 Slots, 10 Poles, Single Layer	11
5	Machine Sizing – The Output Equation	12
5.1	Magnetic and Electric Loading	12
5.2	Output Equation	12
5.3	Worked Example – Pyrhönen 6.1: 4 kW Induction Motor	13
5.4	Aspect Ratio	13
5.5	Worked Example – 30 kW Squirrel-Cage Induction Generator	13
6	Mechanical Constraints	14
6.1	Air-Gap Selection	14
6.2	Tip Speed and Centrifugal Stress	14
6.3	Slot-Height Optimisation	15
7	Main Flux Paths	15
7.1	Carter’s Coefficient	15
7.2	Effective Core Length	16
8	Magnetizing Inductance	16
8.1	Flux per Pole	16
8.2	MMF of a Polyphase Winding	16
8.3	L_m Derivation	17
9	Leakage Flux	17
9.1	Slot Leakage	17
9.2	End-Winding Leakage	18
9.3	Magnet-to-Magnet and Tooth-Tip Leakage	18
9.4	Cogging Torque	18

10 Thermal Modelling	18
10.1 Electrical–Thermal Analogy	18
10.2 Conduction	19
10.3 Convection	19
10.4 Worked Example – Thermal Uprate	19
10.5 Thermal Network for a Machine	19
11 Permanent Magnet Machines	20
11.1 Hard vs. Soft Ferromagnetic Materials	20
11.2 Magnet Equivalent Circuit	20
11.3 Operating Point and Sizing Intuition	20
11.4 Demagnetisation Safety	21
A Summary of Key Formulas	21
B Symbols and Units	22

1 Magnetic Fundamentals

1.1 Electric–Magnetic Analogy

A magnetic circuit obeys the same network laws as an electric circuit once the analogous quantities are identified (Table 1). The driving quantity for current in the electric circuit is the voltage V ; in the magnetic circuit it is the magnetomotive force (MMF) NI , the product of the coil turns and the current they carry. The opposing quantity is the resistance R on the electric side and the reluctance $\mathcal{R}$ on the magnetic side. The resulting flow quantities are current I and flux Φ :

$$V = IR \quad \Longleftrightarrow \quad NI = \Phi \mathcal{R} \quad (1)$$

Table 1: Analogous quantities in electric and magnetic circuits.

Quantity (electric)	(magnetic)	Definition
Voltage V [V]	MMF NI [A·t]	driving source
Current I [A]	Flux Φ [Wb]	flow quantity
Current density $J = \sigma E$ [A/m ²]	Flux density $B = \mu H$ [Wb/m ² = T]	flow per area
Field E [V/m]	Field H [A/m]	gradient of source
Conductivity σ	Permeability μ	material property
Resistance $R = \ell/(\sigma A)$	Reluctance $\mathcal{R} = \ell/(\mu A)$	opposes flow

For a series magnetic path of n segments, $\mathcal{R}_{\text{tot}} = \sum_i \ell_i/(\mu_i A_i)$. The inductance of an N -turn coil wound on a magnetic circuit of total reluctance $\mathcal{R}$ is

$$L = \frac{\lambda}{I} = \frac{N\Phi}{I} = \frac{N \cdot NI/\mathcal{R}}{I} = \frac{N^2}{\mathcal{R}} \quad (2)$$

1.2 Air is a Circuit Element

A magnetic circuit cannot be drawn the way an electric circuit can, because flux always closes on itself *including through air*. Whenever the path through a high-permeability core encounters a gap, the air gap appears in series with a large reluctance $\mathcal{R}_g = g/(\mu_0 A)$. Since μ_{Fe}/μ_0 is typically 10^3 – 10^4 , even a millimetre-thick gap dominates the total reluctance of a machine magnetic circuit. This is the central reason why electrical machines are sized by their air-gap geometry rather than by their iron geometry.

1.3 B–H Curves and Permeability

Magnetic materials are non-linear: $B(H)$ saturates and follows a hysteresis loop on reversal. Two different “permeabilities” are commonly quoted (Figure 1):

- **Apparent permeability** $\mu = B/H$ – the secant slope from the origin to the operating point. Used when the material is treated as linear at one operating point.

- **Incremental permeability** $\mu = dB/dH$ – the local tangent slope. Relevant for small-signal AC excitation superimposed on a DC bias.

<i>B-H curve, single-valued</i> (linear-material approximation; secant and tangent slopes shown).	<i>Hysteresis loop</i> (coercive force H_c ; loop area = hysteresis loss per cycle).
--	---

Figure 1: Magnetic material behaviour. Hysteresis loss is independent of frequency per cycle, so the total loss scales linearly with frequency.

Coercive force and hard vs soft materials. The coercive force H_c is the value of H required to drive B back to zero from remanence. Materials with small H_c are easily demagnetised and are called *soft* (used for cores); materials with large H_c are hard to demagnetise and are called *hard* (used for permanent magnets). The product $|B||H|$ in the second quadrant defines the *maximum energy product* – the figure of merit for a permanent magnet (Section 11).

1.4 Core Losses

Two mechanisms convert magnetic flux variation into heat in ferromagnetic cores:

1. **Hysteresis loss.** Loop area per cycle; loss per second $\propto f$ (independent of the magnetic properties per cycle, related to the cyclic reversal of domain orientation). For a 50 Hz machine the loop traverses 50 times per second.
2. **Eddy-current loss.** Induced EMFs in the laminations circulate currents that dissipate in the core resistance.

Eddy-current loss derivation. A small loop of resistance R in the laminating direction sees the induced EMF $e = N d\Phi/dt$. If the flux is sinusoidal with peak $\hat{B}$ and frequency f : $e \propto \hat{B} f$, so $V \propto \hat{B} f$ and $P_{\text{eddy}} = V^2/R \propto \hat{B}^2 f^2$. Including the lamination conductivity σ and thickness explicitly,

$$P_{\text{eddy}} \propto \sigma \hat{B}^2 f^2 \quad (3)$$

Lamination of the core breaks the loop into many small parallel sheets, raising the effective R and reducing P_{eddy} .

Total core loss. $P_{\text{core}} = P_{\text{hyst}} + P_{\text{eddy}}$. At 50 Hz this is dominated by hysteresis; at high frequency (PWM-driven motors) eddy losses dominate.

1.5 Faraday's Law and Inductance

The induced voltage on an N -turn coil is

$$e = -N \frac{d\Phi}{dt} = -\frac{d\lambda}{dt} = -\frac{d(N\Phi)}{dt} \quad (4)$$

where $\lambda = N\Phi$ is the flux linkage. The inductance is defined as

$$L = \frac{d\lambda}{dI} = N \frac{d\Phi}{dI} \quad (5)$$

For a *linear* material with constant $\mathcal{R}$, $\Phi = NI/\mathcal{R}$ so $L = N^2/\mathcal{R}$ as in Eq. (2). For a non-linear material with saturation, $\lambda(I)$ is a curve that bends over (Figure 2); $L = \lambda/I$ is then *not* a constant and the value $\frac{1}{2}LI^2$ is *not* the stored energy.

λ - I *curve* (linear region near origin, saturating at high I ; $L = \lambda/I$ only valid in the linear region).

Figure 2: Flux linkage as a function of current.

1.6 Non-Uniform Flux, Fringing and Leakage

Real magnetic circuits depart from the ideal in several ways:

- **Fringing flux** bulges outward in the air gap; the effective gap cross-section is larger than the iron face.
- **Leakage flux** bypasses the intended path (slot-leakage, end-winding leakage, magnet-to-magnet leakage).
- Non-uniform flux density across a section can arise because the inner part of a curved iron path is shorter and thus has lower reluctance, attracting more flux ($B_{\text{in}} > B_{\text{out}}$).

These effects are accounted for either by Carter's coefficient (Section 7) or by an explicit reluctance network.

1.7 Energy Storage in the Magnetic Field

For a linear material the energy density is

$$w = \int H dB = \frac{1}{2} BH = \frac{B^2}{2\mu} \quad (6)$$

and the total stored energy in a volume V is

$$W = \frac{1}{2} LI^2 = \frac{1}{2} \frac{B^2}{\mu} V \quad (7)$$

At fixed B , low μ stores more energy per unit volume; this is why an air gap is added when a large amount of energy must be stored (inductors, transformers in flyback mode, etc.). Conversely, for *zero-energy* flux paths (the iron in a machine) high μ minimises the stored field energy and the magnetising current.

Maxwell stress. The tangential stress on an iron surface in a normal field B is

$$\sigma_{\text{Maxwell}} = \frac{B^2}{2\mu_0} \quad [\text{N/m}^2 = \text{Pa}] \quad (8)$$

This is the same as the energy density of Eq. (6); the units check because $\text{J/m}^3 = \text{N/m}^2$. The integral of this stress over the air-gap surface times the rotor radius gives the electromagnetic torque (Section 5).

2 AC Machine Equivalent Circuit

The per-phase steady-state equivalent circuit of an AC machine (induction motor with the slip-dependent rotor branch as an example) is shown in Figure 3. The stator branch has $R_s + jX_s$, where $X_s = \omega L_{s\sigma}$ is the stator leakage reactance. The shunt branch carries the magnetising current and the iron losses: R_c in parallel with $jX_m = j\omega L_m$. The rotor branch (referred to the stator) is $R'_r + jX'_r$, with the “mechanical resistance” $R'_r(1-s)/s$ representing the converted mechanical power.

Per-phase equivalent circuit: $X_s, R_s - [R_c \parallel jX_m] - X'_r, R'_r, R'_r(1-s)/s$.

Figure 3: Per-phase steady-state equivalent circuit of an induction machine.

Leakage inductance vs. magnetising inductance. The leakage path is partly air, so $L_{s\sigma}$ is closer to a linear inductor (small saturation) and is dominated by the slot/end-winding geometry rather than the iron. The magnetising inductance L_m is set by the air gap and the iron and saturates with I_m .

3 Integral-Slot Distributed Windings

A polyphase winding placed in slots produces an MMF wave that travels around the air gap. The quality of this wave – how close it is to a pure sinusoid, how large the fundamental component is relative to the peak ampere-conductor count – is described by the *winding factor* k_w .

3.1 Slots per Pole per Phase

For a machine with N_s stator slots, m phases and $2p$ poles,

$$q = \frac{N_s}{2pm} \quad (9)$$

is the number of slots per pole per phase. When q is a positive integer, the winding is *integral-slot* (this section); when q is a non-integer rational, the winding is *fractional-slot* (Section 4).

The slot pitch in electrical degrees is

$$\alpha_e = \frac{p \cdot 360^\circ}{N_s} \quad (10)$$

Worked example – 6-slot, 2-pole, 3-phase machine. Here $p = 1$, $m = 3$, $N_s = 6$, so $q = 1$ and $\alpha_e = 60^\circ$. The phase assignment to the six slots is

Slot	1	2	3	4	5	6
Phase	A+	C−	B+	A−	C+	B−

which yields a 60° -sector pattern repeating every pole-pair.

3.2 MMF of a Concentrated Coil

A single-slot full-pitch coil produces a rectangular MMF wave of amplitude $NI/2$ on each side of the magnetic axis, switching sign at $\pm\pi$ (electrical). The Fourier expansion of this square wave is

$$\text{MMF}(\theta) = \frac{4}{\pi} \cdot \frac{NI}{2} \sum_{n=1,3,5,\dots} \frac{1}{n} \sin(n\theta) \quad (11)$$

so the fundamental amplitude is

$$\widehat{\text{MMF}}_1 = \frac{4}{\pi} \cdot \frac{NI}{2} \quad (12)$$

A *distributed* winding (multiple coils per phase per pole spread across q adjacent slots) produces a stepped waveform that approaches a sinusoid as q grows, attenuating the high-order harmonics.

3.3 Distribution Factor k_d

When q coils share one pole-phase belt, their individual EMF phasors are offset by α_e each, so the resultant is the vector sum (a chord of the circumscribed polygon) rather than the arithmetic sum:

$$k_d = \frac{\text{vector sum of } q \text{ coil EMFs}}{\text{arithmetic sum}} \quad (13)$$

The geometric-series identity gives

$$k_{d,n} = \frac{\sin(n q \alpha_e/2)}{q \sin(n \alpha_e/2)} \quad (14)$$

For the fundamental ($n = 1$), k_d is just below unity and increases toward 1 as q increases (smoother distribution). For higher harmonics, k_d drops sharply, which is the harmonic-attenuation benefit of distribution.

3.4 Pitch Factor k_p

A *full-pitch* coil has its two sides exactly one pole pitch apart, so the two side EMFs are 180° out of phase and add constructively in the coil loop. A *short-pitch* coil has its sides closer than one pole pitch by some fraction λ of the pole pitch, $0 < \lambda \leq \pi$ (electrical). The vector subtraction of the two side EMFs has magnitude $2\hat{E} \sin(\lambda/2)$ instead of $2\hat{E}$:

$$k_{p,n} = \sin\left(\frac{n \lambda}{2}\right) \quad (15)$$

For the fundamental, $k_p \leq 1$. At $\lambda = \pi(1 - 1/n)$, $k_{p,n} = 0$ and the n -th harmonic is eliminated entirely. Common choices:

- $\lambda = 5/6 \pi$ (150° , “5/6 pitched”) – suppresses the 11th and 13th in a three-phase machine.
- $\lambda = 2/3 \pi$ (120°) – eliminates the third harmonic (and triplens).
- $\lambda = 4/5 \pi$ (144°) – eliminates the fifth.

Why eliminate harmonics. Space harmonics of the MMF do not contribute average torque (only the fundamental does), but they produce iron losses, torque ripple, acoustic noise, and parasitic asynchronous torques. Even harmonics imply DC bias in the field distribution and are absent by construction in a balanced winding (each half of the loop sees both $+$ and $-$ poles).

3.5 Winding Factor k_w

The total winding factor combines both effects:

$$k_{w,n} = k_{d,n} \cdot k_{p,n} \quad (16)$$

A high $k_{w,1}$ maximises torque density; a low $k_{w,n}$ for $n > 1$ reduces harmonic content.

3.6 Worked Example – 180-slot, 50 Hz, 10-Pole Alternator

A rotating field produced by a balanced 3-phase 50 Hz, 600 rpm alternator has the magnetic flux distribution

$$B(\theta) = 0.9 \sin \theta + 0.25 \sin 3\theta + 0.18 \sin 5\theta \quad (17)$$

The machine has 180 slots, double-layer Y-connected winding, each coil of 3 turns and coil span 15 slots. Armature diameter $D = 1.35$ m, effective axial length $L = 0.5$ m.

Number of poles. $f_m = 600/60 = 10$ Hz, $f_e = 50$ Hz, so $p = f_e/f_m = 5$ pole-pairs (10 poles).

Slots per pole per phase. $q = 180/(3 \cdot 10) = 6$.

Pole area.

$$A_{\text{pole}} = \frac{\pi D L}{2p} = \frac{\pi \cdot 1.35 \cdot 0.5}{10} = 0.212 \text{ m}^2 \quad (18)$$

Slot pitch (electrical) and coil pitch. $\alpha_e = (360^\circ/180) \cdot p = 10^\circ$ electrical; coil span 15 slots $\Rightarrow \lambda = 150^\circ$ electrical.

Distribution factors.

$$\begin{aligned} k_{d,1} &= \sin(6 \cdot 10^\circ/2)/(6 \sin(10^\circ/2)) = \sin 30^\circ/(6 \sin 5^\circ) = 0.956 \\ k_{d,3} &= \sin 90^\circ/(6 \sin 15^\circ) = 0.643 \\ k_{d,5} &= \sin 150^\circ/(6 \sin 25^\circ) = 0.197 \end{aligned}$$

Pitch factors.

$$\begin{aligned} k_{p,1} &= \sin(150^\circ/2) = \sin 75^\circ = 0.966 \\ k_{p,3} &= \sin 225^\circ = -0.707 \\ k_{p,5} &= \sin 375^\circ = \sin 15^\circ = 0.259 \end{aligned}$$

Winding factors.

$$k_{w,1} = 0.923, \quad k_{w,3} = -0.455, \quad k_{w,5} = 0.051 \quad (19)$$

Peak flux per pole (fundamental).

$$\hat{\Phi}_1 = \frac{2}{\pi} \hat{B}_1 A_{\text{pole}} = \frac{2}{\pi} \cdot 0.9 \cdot 0.212 = 0.1214 \text{ Wb} = 121.4 \text{ mWb} \quad (20)$$

Turns per phase. For $q = 6$, $2p = 10$, 3 turns per coil, double-layer: $N_s = q \cdot 2p \cdot 3 = 180$ turns per phase.

Induced phase voltage (per harmonic) by Faraday. The classical alternator formula is

$$V_{\text{ind,phase}} = \frac{2\pi}{\sqrt{2}} f N_s \hat{B} A_{\text{pole}} k_{w,n} = 4.44 f N_s \hat{B} A_{\text{pole}} k_{w,n} \quad (21)$$

For the three harmonics:

$$V_1 \approx 4480 \text{ V}, \quad V_3 \approx -613 \text{ V}, \quad V_5 \approx 49.5 \text{ V (rms)}$$

The phase voltage is the RMS sum $V_{\text{ph}} = \sqrt{V_1^2 + V_3^2 + V_5^2}$. The Y-connection eliminates the third harmonic from the line-to-line voltage (triplens are in phase across the three phases), so the *line* voltage waveform is much closer to sinusoidal than the phase voltage.

4 Fractional-Slot Windings

A *fractional-slot* winding has $q \notin \mathbb{Z}$. Such windings are common in low-speed, high-pole-count machines (direct-drive PM motors, in particular). Their main practical advantages are:

- Short end-windings (when $q < 1$, tooth-coil), lower copper volume and lower DC resistance.
- Reduction of cogging torque, because the slot/pole symmetry is broken.
- Better stator-winding fault tolerance (faulty coils can be isolated without unbalancing the rotating field).

Their main disadvantages are:

- Lower fundamental winding factor (in general) than a comparable distributed winding.
- Rich harmonic content of the MMF, including *sub*-harmonics that can saturate the rotor back-iron of PM machines and cause additional rotor losses.
- More complex analysis: closed-form k_d , k_p no longer apply; the star-of-slots method is needed instead.

4.1 Star of Slots

Each slot k is assigned an EMF phasor at electrical angle

$$\varphi_k = (k - 1) \alpha_s \pmod{360^\circ}, \quad \alpha_s = \frac{p \cdot 360^\circ}{N_s} \quad (22)$$

The N_s phasors form a star (or a regular polygon traced multiple times, depending on $\text{gcd}(N_s, p)$). For a balanced three-phase winding the 360° circle is divided into six 60° sectors:

A+: $[0^\circ, 60^\circ)$; C–: $[60^\circ, 120^\circ)$; B+: $[120^\circ, 180^\circ)$; A–: $[180^\circ, 240^\circ)$; C+: $[240^\circ, 300^\circ)$; B–: $[300^\circ, 360^\circ)$

Each slot is assigned to the phase whose sector contains its phasor. The n -th harmonic winding factor is then computed by the phasor sum

$$k_{w,n} = \frac{1}{N_{ph}} \left| \sum_{k \in \text{ph}^+} e^{jn\varphi_k} + \sum_{k \in \text{ph}^-} e^{j(n\varphi_k + \pi)} \right| \quad (23)$$

where N_{ph} is the number of slots per phase and ph^+/ph^- are the positive and negative slots of that phase.

4.2 Balancing Condition

For a three-phase winding to be balanced, the number of slots per phase N_s/m must be an integer:

$$\frac{N_s}{m} \in \mathbb{Z}^+ \quad (24)$$

This is necessary but not sufficient; in addition, $\gcd(N_s, 2p)$ should be such that the phasor stars of the three phases are identical apart from a 120° rotation. For example:

- 10-pole, 30 slots: $q = 1$, integral, balanced.
- 10-pole, 60 slots: $q = 2$, integral, balanced.
- 10-pole, 42 slots: $q = 7/5$, fractional; $42/3 = 14$ slots/phase is integer, balanced.
- 10-pole, 44 slots: $44/3$ is not integer – not balanced.

4.3 Slot/Pole Combinations

Some slot/pole combinations have particularly high fundamental winding factor and low cogging torque; tabulated examples (see Pyrhönen Table 2.5) include

- 12-slot, 10-pole ($k_{w,1} \approx 0.933$);
- 12-slot, 14-pole ($k_{w,1} \approx 0.933$);
- 15-slot, 14-pole;
- 18-slot, 16-pole.

Multiples of these basic units (24/20, 24/28, ...) are also widely used.

4.4 Worked Example – 12 Slots, 10 Poles, Single Layer

$\alpha_s = (360^\circ/12) \cdot 5 = 150^\circ$ electrical. Drawing the 12 phasors on the star and grouping them by 60° sector yields 4 slots per phase (overlapping pairs at the same angle). The fundamental winding factor obtained from the phasor sum is $k_{w,1} \approx 0.933$. This combination uses tooth-coil (concentrated) windings – each coil is wound around one stator tooth, dramatically shortening the end winding.

5 Machine Sizing – The Output Equation

The classical *sizing equation* relates the apparent power delivered by an AC machine to its airgap volume, the electric loading, the magnetic loading, and the synchronous speed.

5.1 Magnetic and Electric Loading

Specific magnetic loading $\bar{B}$. The average air-gap flux density over one pole (one pole pitch on the bore circumference):

$$\bar{B} = \frac{p \Phi_p}{\pi D L} \quad [\text{T}] \quad (25)$$

where Φ_p is the flux per pole, D the bore diameter and L the axial length. Typical values: 0.6–0.9 T for industrial induction motors, 0.8–1.0 T for PM motors.

Specific electric loading $\bar{A}$. The RMS ampere-turns per unit length of air-gap circumference:

$$\bar{A} = \frac{N_{\text{turn,slot}} I Q}{\pi D_i} \quad [\text{A/m}] \quad (26)$$

where $Q = N_s$ is the slot count, $N_{\text{turn,slot}}$ the turns per slot, I the RMS phase current. Typical values: 20–40 kA/m for industrial motors, up to 80–100 kA/m for forced-cooled machines. Higher $\bar{A}$ raises torque density but worsens $I^2 R$ losses and demands better cooling.

5.2 Output Equation

The apparent power of an m -phase machine is $S = m E_m I_s$ where E_m is the phase RMS EMF and I_s the RMS current. Substituting the alternator formula $E_m = (2\pi/\sqrt{2}) f N_s k_{w,1} \hat{B} A_{\text{pole}}/p$ and grouping with the geometric variables:

$$S_i = \frac{\pi^2}{\sqrt{2}} n_{\text{syn}} k_{w,1} \bar{A} \hat{B} D^2 L \quad (27)$$

The result is usually written

$$\boxed{S_i = C D^2 L n_{\text{syn}}} \quad C = \frac{\pi^2}{\sqrt{2}} k_{w,1} \bar{A} \hat{B} \quad (28)$$

where C is the *specific machine constant*, a single figure of merit that captures how well the machine uses its volume. C depends only on the materials and the cooling capability; it lets machines of different sizes be compared at a glance.

How to increase power. From Eq. (27), the three independent levers are:

1. Increase the volume ($D^2 L$): build a physically larger machine.
2. Increase the loadings ($\bar{A}$, $\hat{B}$): better materials, better cooling, higher current density.
3. Increase n_{syn} : spin it faster.

The number of poles by itself does *not* change S_i ; for a fixed n_{syn} , more poles means higher f but the same output power.

5.3 Worked Example – Pyrhönen 6.1: 4 kW Induction Motor

Given $D = 98 \text{ mm}$, $L = 82 \text{ mm}$, $P = 4 \text{ kW}$, $f = 50 \text{ Hz}$, $2p = 2$, rated slip $s = 0.01$.

Synchronous speed $n_{\text{syn}} = f/p = 50 \text{ Hz} = 3000 \text{ rpm}$, mechanical $\omega_{\text{mech}} = 2\pi f(1 - s) = 99\pi \text{ rad/s}$.

Torque:

$$T = \frac{P}{\omega_{\text{mech}}} = \frac{4000}{99\pi} \approx 12.86 \text{ Nm} \quad (29)$$

Air-gap (rotor surface) area: $S_r = \pi DL = \pi \cdot 0.098 \cdot 0.082 = 0.0252 \text{ m}^2$.

Tangential force and shear stress:

$$F_{\text{tan}} = T/r = 12.86/0.049 = 262 \text{ N}, \quad \sigma = F_{\text{tan}}/S_r = 262/0.0252 \approx 10.4 \text{ kPa} \quad (30)$$

This is in the standard 5–35 kPa band for industrial non-PM motors.

Specific machine constant (mechanical):

$$C_{\text{mech}} = \frac{P}{D^2 L f_{\text{sync}}} = \frac{4000}{0.098^2 \cdot 0.082 \cdot 50} \approx 102 \text{ kW} \cdot \text{s/m}^3 \quad (31)$$

5.4 Aspect Ratio

The ratio $\chi = L/D$ is a free design parameter constrained mostly by mechanical and thermal considerations. For asynchronous machines a common rule of thumb is

$$\chi \approx \frac{\pi}{2p} \sqrt[3]{p} \quad (32)$$

with $0.5 \lesssim D/L \lesssim 2.5$ as a typical industrial band. Synchronous machines often have $\chi < 1$ (pancake) for direct-drive applications and $\chi > 1$ (slender) for high-speed applications.

5.5 Worked Example – 30 kW Squirrel-Cage Induction Generator

A 30 kW, 690 V, 50 Hz, 4-pole, 3-phase squirrel-cage induction generator is to be sized. Choose $C_{\text{mech}} = 150 \text{ kW/m}^3$ (a typical value for a totally-enclosed fan-cooled machine).

Aspect ratio:

$$\chi \approx \frac{\pi \sqrt[3]{p}}{2p} = \frac{\pi \sqrt[3]{2}}{2 \cdot 2} = 0.989 \approx 1 \quad (33)$$

i.e. $L \approx D$, so $D^2 L \approx D^3$.

Bore diameter:

$$D = \sqrt[3]{\frac{P}{C_{\text{mech}} f_{\text{sync}}}} = \sqrt[3]{\frac{30\,000}{150\,000 \cdot 25}} \approx 200 \text{ mm} \quad (34)$$

(taking $f_{\text{sync}} = f/p = 25 \text{ rev/s}$ for the 4-pole machine). With $L \approx D = 200 \text{ mm}$.

Mechanical speed $n_{\text{mech}} \approx 1474 \text{ rpm}$ (slip $\sim 1.7\%$), so the torque is $T = P/\omega = 30\,000/(2\pi \cdot 1474/60) \approx 194.4 \text{ Nm}$.

Tangential stress at the rotor surface:

$$\sigma_{\text{tan}} = \frac{T}{S_r r} = \frac{T}{\pi D L \cdot D/2} = \frac{194.4}{\pi(0.2)^3/2} \approx 15.5 \text{ kPa} \quad (35)$$

again within the standard industrial band.

Stator circumference $\pi D = 628 \text{ mm}$. For $q = 2$: $N_s = 2 \cdot 4 \cdot 3 = 24$ slots; assuming equal slot and tooth widths, the tooth thickness is $\tau_{\text{teeth}} = 628/(2 \cdot 24) = 13 \text{ mm}$. For $q = 4$: $N_s = 48$, $\tau_{\text{teeth}} = 6.5 \text{ mm}$. For $q = 3$: $N_s = 36$, $\tau_{\text{teeth}} = 8 \text{ mm}$ – a reasonable compromise. Coil span: about 9 slots.

6 Mechanical Constraints

6.1 Air-Gap Selection

The air gap g is set by mechanical clearance, eccentricity tolerance, and the level of additional losses one is willing to accept from slot-harmonic field penetration. An empirical rule (Pyrhönen Eq. 6.7) for industrial induction motors:

$$g \approx 1.6 \times 0.18 \times 0.006 \times P^{0.4} \quad [\text{m}, \text{W}] \quad (36)$$

For the 30 kW machine above this gives $g \approx 0.988 \text{ mm}$. Heavy-duty service typically warrants a 60% increase ($g \approx 1.5 \text{ mm}$ here). PM machines often use the smallest mechanically feasible gap to minimise the required magnet volume.

6.2 Tip Speed and Centrifugal Stress

The tangential (tip) speed of the rotor is $v = \omega r$. For a 0.25 m radius rotor to reach Mach 1 ($v \approx 345 \text{ m/s}$) the angular speed is $\omega = v/r \approx 1400 \text{ rad/s}$, i.e. $n \approx 14\,000 \text{ rpm}$ – not feasible for any reasonable rotor.

The centrifugal radial stress in a smooth solid rotor of density ρ at the surface is

$$\sigma_{\text{mech}} = C' \rho \omega^2 r^2, \quad C' = \frac{3 + \nu}{4} \quad (37)$$

where ν is Poisson's ratio. For steel, $\nu \approx 0.29$, so $C' \approx 0.823$.

Worked example – Pyrhönen 6.3. A smooth steel cylinder with a small bore, $\rho = 7860 \text{ kg/m}^3$, yield strength $\sigma_y = 300 \text{ MPa}$, $\nu = 0.29$, spinning at $n = 15\,000 \text{ rpm}$.

$$r_{\text{max}} = \sqrt{\frac{\sigma_y}{C' \rho \omega^2}} = \sqrt{\frac{300 \cdot 10^6}{0.823 \cdot 7860 \cdot (2\pi \cdot 15000/60)^2}} \approx 0.13 \text{ m} \quad (38)$$

For comparison, a 0.25 m radius at the same speed would *exceed* the yield strength – this is the fundamental reason that high-speed machines must be slender (and why the aspect ratio χ grows for high-speed designs).

6.3 Slot-Height Optimisation

Tall slots offer more area for copper, allowing higher electric loading $\bar{A}$. But there are tradeoffs:

- Cooling becomes harder (heat has to travel further out of the deep slot).
- Slot leakage flux grows.
- If D_{rotor} is held constant, the outer diameter grows; if D_{out} is held constant, the rotor shrinks and the torque drops because $T \propto D^2 L$.

A standard optimisation argument shows that the optimum normalised slot height $d = h_{\text{slot}} / (R_{\text{out}} - R_{\text{rotor}})$ is about 0.5: torque scales as $T \propto (1 - d)d$, maximised at $d = 1/2$. A common practical range is $0.3 \leq d \leq 0.6$.

7 Main Flux Paths

7.1 Carter's Coefficient

Stator slotting causes the air-gap flux density to dip above each slot opening, raising the effective reluctance of the gap. Carter's coefficient $k_c \geq 1$ replaces the slotted stator with a smooth stator of enlarged effective gap:

$$g_{\text{eff}} = k_c g \quad (39)$$

The simplified Pyrhönen form (Eq. 3.34) is

$$k_c = \frac{\tau_u}{\tau_u - b_e}, \quad b_e = \kappa b_1, \quad \kappa = \frac{b_1/g}{5 + b_1/g} \quad (40)$$

with τ_u the slot pitch at the bore, b_1 the slot opening, and g the physical gap. $\kappa \rightarrow 0$ for small b_1/g (no slotting effect) and $\kappa \rightarrow 1$ for very wide open slots.

Worked example – Pyrhönen 3.1. $g = 0.8 \text{ mm}$, $b_1 = 3 \text{ mm}$, $\tau_u = 10 \text{ mm}$.

$$\begin{aligned} \kappa &= \frac{b_1/g}{5 + b_1/g} = \frac{3/0.8}{5 + 3/0.8} = \frac{3.75}{8.75} = 0.429 \\ b_e &= 0.429 \cdot 3 = 1.29 \text{ mm} \\ k_c &= \frac{10}{10 - 1.29} = 1.148 \\ g_{\text{eff}} &= 0.8 \cdot 1.148 = 0.918 \text{ mm} \end{aligned}$$

Armature reaction. In addition to the magnet (or rotor) field, the stator current sheet creates a field of its own that adds to the no-load gap field. For a linear core the armature reaction merely shifts the operating point; for a non-linear core close to saturation the reaction effect can dominate, distorting the field shape further.

7.2 Effective Core Length

The axial extent of a real stator stack is broken up by cooling ducts between packs of laminations. Flux fringes from one pack to the next across each duct, so the effective magnetic length L' is somewhat greater than the iron length and somewhat less than the total stack length.

Worked example – Pyrhönen 3.2. 25 stator sectors of 30 mm each separated by 24 cooling ducts of $b_v = 10$ mm each. The total stack length is $L = 25 \cdot 30 + 24 \cdot 10 = 990$ mm (the iron length alone is 750 mm). With air gap $g = 3$ mm:

$$\kappa_v = \frac{b_v/g}{5 + b_v/g} = \frac{10/3}{5 + 10/3} = 0.4, \quad b_{ve} = 0.4 \cdot 10 = 4 \text{ mm} \quad (41)$$

The effective core length is then approximately

$$L' \approx L - n_v b_{ve} + 2g = 990 - 24 \cdot 4 + 2 \cdot 3 = 900 \text{ mm} \quad (42)$$

(the $+2g$ term accounts for axial-end fringing).

8 Magnetizing Inductance

The magnetizing inductance L_m of an AC machine is the self-inductance of one phase due to the fundamental-component air-gap field it creates when carrying only its own current and no saturating component. It is the dominant element of the shunt branch in the per-phase equivalent circuit (Section 2).

8.1 Flux per Pole

A sinusoidally distributed air-gap flux density of peak value $\hat{B}$ integrated over half a wavelength (one pole pitch τ_p , on axial length L') gives the peak flux per pole:

$$\hat{\Phi}_{\text{pole}} = \int_0^\pi \hat{B} \sin \theta \cdot \frac{\tau_p}{\pi} L' d\theta = \frac{2}{\pi} \hat{B} \tau_p L' \quad (43)$$

The corresponding stator flux linkage is $\lambda = k_{w,1} N_s \hat{\Phi}_{\text{pole}}$.

8.2 MMF of a Polyphase Winding

For a balanced m -phase winding with N_s turns per phase, the fundamental space MMF amplitude is

$$\widehat{\text{MMF}}_1 = \frac{4}{\pi} \frac{k_{w,1} N_s}{2p} \sqrt{2} I_{s,\text{rms}} \quad (44)$$

The factor $\frac{4}{\pi}$ is the Fourier amplitude of a square-wave ampere-turn distribution; division by $2p$ converts total turns to turns per pole; the $\sqrt{2}$ converts RMS to peak.

8.3 L_m Derivation

Setting the MMF equal to $\hat{H} g_{\text{eff}} = \hat{B} g_{\text{eff}}/\mu_0$ and solving for $\hat{B}$:

$$\hat{B} = \frac{\mu_0}{g_{\text{eff}}} \cdot \frac{4}{\pi} \frac{k_{w,1} N_s}{2p} \sqrt{2} I_{s,\text{rms}} \quad (45)$$

The flux linkage per phase becomes

$$\lambda = \frac{2}{\pi} \frac{\mu_0}{g_{\text{eff}}} \frac{4}{\pi} \frac{(k_{w,1} N_s)^2}{2p} \tau_p L' \sqrt{2} I_{s,\text{rms}} \quad (46)$$

Dividing by $\sqrt{2} I_{s,\text{rms}}$ gives the per-phase magnetizing inductance:

$$L_m^{(\text{ph})} = \frac{2\mu_0 D}{p^2 \pi g_{\text{eff}}} L' (k_{w,1} N_s)^2 \quad (47)$$

where $\tau_p = \pi D/(2p)$ has been used. For three-phase analysis one usually quotes the *cyclic* magnetizing inductance

$$L_m = \frac{3}{2} L_m^{(\text{ph})} \quad (48)$$

which is the inductance seen in the equivalent dq circuit.

Induction motor implications. $L_m \propto 1/g_{\text{eff}}$. A larger effective gap reduces L_m , increases the magnetising current I_m , lowers the power factor, and ultimately reduces efficiency. This is why induction motors are designed with the smallest mechanically feasible gap (typically a fraction of a millimetre for sub-kW motors, a few millimetres at multi-MW).

9 Leakage Flux

Flux that links *some but not all* of the winding turns is called leakage flux. It does not contribute to torque, but it does store energy and shows up in the equivalent circuit as the leakage reactance.

9.1 Slot Leakage

Flux closing around a single tooth slot without linking the rotor winding constitutes slot leakage. Applying Ampère's law on a closed contour inside the slot,

$$\oint \vec{H} \cdot d\vec{\ell} = NI, \quad (49)$$

the contour can be split into a vertical H_2 portion (close to the slot bottom, with most of the enclosed turns) and a horizontal H_1 portion (across the slot mouth, with few turns enclosed). Since $H_2 \gg H_1$ but $\ell_2 \ll \ell_1$, the dominant contribution is the slot-bottom path; this is the source of the large slot-leakage inductance.

9.2 End-Winding Leakage

End-windings carry the full phase current but do not contribute to the useful air-gap field. The associated leakage inductance is roughly proportional to the end-winding length and is one of the reasons why fractional-slot tooth-coil windings (with very short end turns) achieve high torque density.

9.3 Magnet-to-Magnet and Tooth-Tip Leakage

In PM machines, a fraction of the magnet flux closes from one magnet to its neighbour through the air at the magnet ends, and through the tooth tips back to the stator slots. These leakage paths can be modelled with a reluctance network and are responsible for the reduction of the useful air-gap fundamental below the value computed from a 1D MEC.

9.4 Cogging Torque

A static interaction between the magnets and the stator slot openings produces a position-dependent reluctance, hence a position-dependent torque even at zero stator current. This *cogging torque* is a torque ripple, not useful for propulsion. Fractional-slot windings attenuate it; further reduction comes from skewing the stator or the magnets by one slot pitch.

10 Thermal Modelling

10.1 Electrical–Thermal Analogy

Heat flow through a solid obeys the same diffusion equation as electric current in a resistor:

$$q = \frac{\Delta T}{R_{\text{th}}} \quad \Longleftrightarrow \quad I = \frac{\Delta V}{R} \quad (50)$$

The analogy is summarised in Table 2.

Table 2: Electrical–thermal analogy.

Electrical	Thermal
Voltage V [V]	Temperature T [K]
Current I [A]	Heat flow q [W]
Resistance R [Ω]	Thermal resistance R_{th} [K/W]
Capacitance C [F]	Heat capacity mc_p [J/K]
Conductivity σ	Thermal conductivity k [W/(m·K)]

A lumped thermal network of resistors and capacitors – one node per component, one resistor per heat-flow path, one capacitor per thermal mass – can predict steady-state and transient temperature rise without solving any PDE.

10.2 Conduction

For a slab of thickness ℓ , area A and conductivity k :

$$R_{\text{th,cond}} = \frac{\ell}{k A} \quad (51)$$

A copper slot, for example, conducts heat radially outward through the slot insulation (low k), the stator tooth iron (medium k), and into the frame.

10.3 Convection

For a surface of area A exposed to a coolant with film coefficient h :

$$R_{\text{th,conv}} = \frac{1}{h A} \quad (52)$$

Natural convection $h \sim 5\text{--}25 \text{ W}/(\text{m}^2\text{K})$; forced air $h \sim 25\text{--}250 \text{ W}/(\text{m}^2\text{K})$; water cooling $h \sim 10^3\text{--}10^4 \text{ W}/(\text{m}^2\text{K})$.

Fan-load scaling. For a fan, the torque scales as $T \propto \omega^2$ and the power as $P = T\omega \propto \omega^3$. Doubling the airflow speed therefore requires eight times the shaft power – which is why uprating the cooling of an existing machine by simply spinning the fan faster has diminishing returns.

10.4 Worked Example – Thermal Uprate

A motor rated $P_{\text{out}} = 10 \text{ kW}$, $\eta = 0.9$, $T_{\text{amb}} = 40^\circ\text{C}$, $T_{\text{motor}} = 80^\circ\text{C}$ at rating. So $P_{\text{loss}} = P_{\text{out}}/\eta - P_{\text{out}} = 1 \text{ kW}$ and $\Delta T = 40 \text{ K}$, giving the steady-state thermal resistance $R_{\text{th}} = 40 \text{ K/kW} = 0.04 \text{ K/W}$.

Uprate by 50%. If the same motor must deliver 15 kW, the I^2R losses scale as $I^2 \propto P^2$, so $P_{\text{loss}}^{\text{new}} = 1 \text{ kW} \cdot (1.5)^2 = 2.25 \text{ kW}$ and the temperature rise becomes $\Delta T = R_{\text{th}} \cdot P_{\text{loss}}^{\text{new}} = 40 \cdot 2.25 = 90 \text{ K}$ above ambient, i.e. $T_{\text{motor}} \approx 130^\circ\text{C}$. A more careful calculation that also lets the winding resistance grow with temperature (a $\sim 25\%$ increase at 130°C relative to 80°C) gives $\Delta T \approx 112.5 \text{ K}$ and $T_{\text{motor}} \approx 152^\circ\text{C}$ – likely beyond Class F insulation limits, so the uprate is not sustainable without improved cooling.

10.5 Thermal Network for a Machine

A practical thermal model groups losses by region (copper, iron, end winding, magnet) and connects them through the local conduction and convection resistances. Heat then flows from copper $\rightarrow$ slot insulation $\rightarrow$ tooth iron $\rightarrow$ back-iron $\rightarrow$ frame $\rightarrow$ ambient. Air ducts add parallel convection paths from the iron to the duct air, in series with the duct-to-ambient resistance. The end-winding region has its own convection path to the internal air volume.

11 Permanent Magnet Machines

11.1 Hard vs. Soft Ferromagnetic Materials

The B–H loop of a *soft* ferromagnetic material (silicon steel, nickel-iron alloys) has small enclosed area and small coercive force H_c : it follows the applied field easily and reverses with little hysteresis loss. Soft materials are used for the stator and rotor cores of all electrical machines.

A *hard* ferromagnetic material (ferrite, samarium-cobalt, neodymium-iron-boron) has very large H_c and large remanent flux density B_r . Once magnetised it retains its magnetisation against substantial demagnetising fields. The figure of merit is the *maximum energy product* $|BH|_{\max}$ – the area of the largest rectangle that fits in the second-quadrant B–H characteristic. Typical values:

- Ferrite: $B_r \approx 0.4$ T, $H_c \approx 250$ kA/m, $|BH|_{\max} \approx 30$ kJ/m³.
- NdFeB: $B_r \approx 1.2$ – 1.4 T, $H_c \approx 800$ – 1200 kA/m, $|BH|_{\max} \approx 250$ – 430 kJ/m³.

11.2 Magnet Equivalent Circuit

In the linear (recoil) region of the second quadrant, the magnet behaves like an MMF source $H_c l_m$ in series with an internal reluctance $\mathcal{R}_m = l_m/(\mu_0 \mu_r A_m)$, where $\mu_r \approx 1.05$ for NdFeB. Equivalently, on the B axis, the source is the remanent flux $\Phi_r = B_r A_m$. The whole magnetic circuit becomes a flux source driving an external reluctance (the air gap, the iron, and any leakage paths) in series with $\mathcal{R}_m$:

$$\Phi_g = \frac{B_r A_m}{1 + \mathcal{R}_g/\mathcal{R}_m} \quad (53)$$

In the idealised 1D case with $A_m = A_g = A$ and infinite iron permeability, this reduces (cf. Section 5 of the HW2 report) to

$$B_g = \frac{B_r l_m}{l_m + \mu_r g} \quad (54)$$

With Carter’s correction, g is replaced by $k_c g$.

11.3 Operating Point and Sizing Intuition

The operating point of the magnet is the intersection of its demagnetisation line with the load line $-B/(\mu_0 H) = \mathcal{R}_m/\mathcal{R}_g$. Moving the operating point closer to $|BH|_{\max}$ – a load line through the “knee” of the recoil curve – maximises the energy delivered per unit volume of magnet material.

Doubling the magnet thickness does not double B_g . The MMF source doubles but the magnet reluctance also doubles, while the air-gap reluctance is unchanged. The increase in B_g is therefore sub-linear: e.g. for $l_m = 4$ mm, $g = 4$ mm the gap field might be $B_g = 0.6$ T; doubling the magnet to $l_m = 8$ mm raises B_g to roughly 0.8 T, not 1.2 T. Conversely, a magnet thinner than $\mu_r g$ is essentially short-circuited by its own reluctance and delivers a small fraction of B_r .

Maximum energy transfer. The classical maximum-power-transfer theorem applies: the energy density delivered to the external circuit is maximised when the external (load) reluctance equals the internal (magnet) reluctance. For a fixed magnet, this fixes the optimum effective air gap; for a fixed air gap, it fixes the optimum magnet thickness:

$$l_m^{\text{opt}} \approx \mu_r g_{\text{eff}}. \quad (55)$$

In practice, the working point is pushed slightly toward the high- B side of this optimum because demagnetisation safety margin is more valuable than the last few percent of efficiency.

11.4 Demagnetisation Safety

The temperature dependence of B_r (typically $-0.1\%/^{\circ}\text{C}$ for NdFeB) and of H_c (typically $-0.5\%/^{\circ}\text{C}$) shifts the demagnetisation line toward the origin as the magnet heats up. A short-circuit fault, an inverter shoot-through, or a starting current transient can drive the operating point below the *knee* of the recoil line; below the knee, recoil is irreversible and the magnet loses some of its B_r permanently. Conservative PM machine design therefore checks the worst-case operating point (highest temperature, largest stator current, demagnetising stator MMF) against a margin to the knee.

A Summary of Key Formulas

- Slots per pole per phase: $q = N_s/(2pm)$.
- Distribution factor: $k_{d,n} = \sin(nq\alpha_e/2)/[q \sin(n\alpha_e/2)]$.
- Pitch factor: $k_{p,n} = \sin(n\lambda/2)$, with λ the coil span in electrical degrees.
- Winding factor: $k_{w,n} = k_{d,n} k_{p,n}$.
- Flux per pole: $\hat{\Phi}_{\text{pole}} = (2/\pi)\hat{B} \tau_p L'$.
- Alternator EMF (per harmonic): $V_{\text{rms}} = 4.44 f N_s k_{w,n} \hat{B} A_{\text{pole}}$.
- Output equation: $S_i = (\pi^2/\sqrt{2}) k_{w,1} \bar{A} \hat{B} D^2 L n_{\text{syn}} = C D^2 L n_{\text{syn}}$.
- Carter's coefficient: $k_c = \tau_u/(\tau_u - b_e)$, $b_e = \kappa b_1$, $\kappa = (b_1/g)/(5 + b_1/g)$.
- Effective core length: $L' \approx L - n_v b_{ve} + 2g$.
- Magnetising inductance per phase: $L_m^{(\text{ph})} = (2\mu_0 D/(p^2 \pi g_{\text{eff}})) L' (k_{w,1} N_s)^2$.
- Shear stress / Maxwell stress: $\sigma = B^2/(2\mu_0)$.
- Air-gap flux density of a surface-PM machine: $B_g = B_r l_m/(l_m + \mu_r k_c g)$.

B Symbols and Units

Symbol	Meaning	Typical unit
D	stator bore diameter	m
L	axial (stack) length	m
L'	effective magnetic length	m
g	physical air-gap length	mm
g_{eff}	effective air-gap (with k_c)	mm
τ_s	slot pitch at the bore	mm
τ_p	pole pitch at the bore	mm
b_1, w_s	slot opening width	mm
w_{tb}	tooth body width	mm
h_{slot}	slot height	mm
N_s	total slot count	—
$2p$	pole count	—
m	phase count	—
q	slots per pole per phase	—
α_e	slot pitch, electrical	deg
k_d, k_p, k_w	winding factors	—
$\bar{B}$	specific magnetic loading	T
$\bar{A}$	specific electric loading	A/m
C	specific machine constant	W·s/m ³
B_r	magnet remanence	T
H_c	magnet coercivity	A/m
μ_r	magnet relative permeability	— (usually ~ 1.05)
l_m	magnet radial thickness	mm